INFORMATION TO KEEP WITH YOU

My complete treatment handbook

Pembrolizumab · every 3 weeks · advanced melanoma

Understand your treatment. Know what to expect. Keep your plan to hand.

Displayed calendar
21 days
See the active treatment phase

Treatment stages
1
Explained in the treatment reference

Regimen source
NCCP 00455
Version 16

Reference treatment cycle

Treatment stages

01 **Ongoing treatment and review**
Every 3 weeks

Your team confirms the active stage and dates. The full treatment guide explains each step.

Inside your handbook

		PAGE
01	Understanding your treatment	2
02	Your schedule, tests and appointments	6
03	Symptoms and supportive care	12
04	Sources and further information	20

My name

Hospital / oncology team

Use the urgent instructions and contact numbers in your treatment guide. Your team must replace example numbers before patient use. Your individual treatment plan takes precedence.

CHAPTER 01

Understanding your treatment

IN THIS CHAPTER

Why this treatment is offered, what its medicines do, and how the agreed plan fits your care.

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Immunotherapy can cause inflammation in healthy organs during treatment or after it stops. Contact the oncology team promptly for new diarrhoea, blood in stools, cough or breathlessness, yellow skin or eyes, severe headache, vision changes, marked thirst, unusual weakness or a spreading rash. Tell any clinician that you are receiving or have received immunotherapy.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

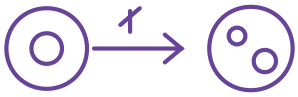
Medicines in this regimen

Pembrolizumab helps immune cells act against melanoma that cannot be completely removed or has spread. The aim is disease control; benefit and side effects are reviewed for you.

PD-1 CHECKPOINT IMMUNOTHERAPY

Pembrolizumab

Keytruda



Releases a PD-1 immune brake

Pembrolizumab blocks the PD-1 checkpoint, allowing T cells to remain more active against cancer. Immune activity can also affect normal organs, so new symptoms matter during treatment and after it stops.

PD-1 blocked → Immune activity increases → Healthy organs can inflame

What to watch for

Carry your immunotherapy alert card and report new symptoms early, including after treatment ends.

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

Understanding your diagnosis and treatment goal

Use these questions with your own pathology report and agreed treatment plan.

What the diagnosis says

Record the original melanoma site and pathology findings. Breslow thickness and ulceration describe features of the skin tumour; stage describes its extent. Melanoma arising in the eye or a mucosal surface may need a different pathway.

BRAF and other tumour tests

BRAF V600 testing can identify an alternative targeted-treatment option in some melanomas. It is separate from the immunotherapy described here. Ask what your tumour result means for the choices available to you.



Surgery, radiotherapy and medicine have different roles

Ask whether medicine is intended to reduce recurrence after complete surgery or control cancer that remains. Add any operation or radiotherapy to the plan with its own purpose, timing and risks.

Stage and response are different

Stage describes the extent of disease. Response describes a change in measurable cancer during treatment. After complete removal there may be no visible tumour to measure; follow-up then looks for recurrence and treatment effects.



A new skin change still needs review

Report a new or changing mole, a sore that persists, a new lump or an unexplained symptom. A photo or a good response in one area cannot rule out a separate problem.

Your medicines and the overall plan

The schedule in this setting

Pembrolizumab every 3 weeks. Treatment continues until progression or unacceptable toxicity. For sustained response without progression, stopping after 24 months may be considered by the treating team; this is not an automatic stop.

The role of earlier or later treatment

Ask how the medicine fits with previous or planned local treatment. The timing of an operation, radiotherapy and a tablet course should be written in one agreed plan.

Pauses and stopping

A safety pause is an active treatment decision. Record which medicine is paused, the reason, the tests or recovery needed and who may authorise restarting. Do not restart from an old prescription or reset the history when a new phase begins.

Questions, decisions and practical support

Your priorities and the burden of treatment belong in the discussion.

Benefits and uncertainties

Ask what outcome this treatment aims to improve for your diagnosis and what alternatives exist. Results from a study group help a discussion but cannot promise your own result.

Alternatives and review points

Agree the next decision point and what would make the plan change. Ask about other suitable treatments, trials and symptom care. Support for comfort and function can continue alongside cancer treatment.

Help with daily life

Tell the team what is hardest in daily life: travel, work, caring duties, costs, sleep or confidence. Ask who can help and how to contact them.

Shared understanding

Before going home, check that you can describe your medicine schedule, next tests and urgent-contact route. Ask for an interpreter, larger print or a supporter if that would help.

What the evidence means for your treatment

Discuss the evidence for your diagnosis, tumour test results and treatment setting.

Evidence in advanced melanoma measures control of existing cancer; adjuvant evidence measures recurrence after surgery. Perioperative evidence concerns treatment around surgery. Ask which study and outcome match your stage and plan. These results cannot predict your individual benefit.

CHAPTER 02

Your schedule, tests and appointments

IN THIS CHAPTER

Your treatment stages, checks and practical preparations, from before treatment through follow-up.

The overall treatment plan

Pembrolizumab · every 3 weeks · advanced melanoma

21-day view



This shows treatment timing, not guaranteed symptom-free days. Your team confirms the active phase and dates.

EVERY 3 WEEKS

Ongoing treatment and review

Day 1 every 21 days **Duration individually reviewed**

Pembrolizumab is given as 200 mg by intravenous (IV) infusion over 30 minutes, on Day 1 every 3 weeks. Allow extra time for assessment, preparation and observation. Treatment continues until progression or unacceptable toxicity. For sustained response without progression, stopping after 24 months may be considered by the treating team; this is not an automatic stop.

The team confirms treatment dates, recovery and each safety decision. A planned treatment count does not authorise the next dose.

BASELINE → TREATMENT → FOLLOW-UP

Required safety checks and your individual disease-assessment plan

The team supplies actual dates and resolves any source ambiguity. A reference period does not replace a prescription, a local monitoring plan or an individual treatment decision.

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00455 v16: current regimen source

THE WHOLE TREATMENT JOURNEY

Pembrolizumab · every 3 weeks · advanced melanoma: treatment and checks

Unresectable or metastatic melanoma in adults; the team confirms the source-specific eligibility and previous treatment.

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

Treatment, tests and reviews across the course

Care / check	Before starting	During treatment	Between visits	Response / follow-up	After a hold or stop
Bloods and organ safety NCCP	Blood count, kidney and liver tests, glucose and thyroid tests; review autoimmune conditions, transplants, previous immune reactions and all current medicines.	Blood count, kidney/liver tests and glucose before every 3-week cycle. thyroid-stimulating hormone (TSH) is checked every 3–6 weeks, including between treatment visits when needed. Cortisol and other tests are added when clinically indicated. Report new symptoms at every visit and between visits; normal blood results do not rule out immune toxicity.	Contact the team for new symptoms; extra tests may be needed between scheduled checks.	Safety bloods do not by themselves show whether cancer is responding.	Continue to report possible immune effects after the last dose; your team records the follow-up tests and dates. Keep any prescribed steroid taper, hormone replacement and specialist follow-up. Scans follow your individual disease plan.
Cancer, skin and daily function NCCP	Record diagnosis, stage, current disease sites and the baseline assessment.	Keep the source-specific skin and clinical reviews and the individually agreed response assessments.	Record new lesions, lumps, wound changes, pain and effects on usual activities.	Write the next examination or scan date and the responsible team; no universal scan or tumour-marker interval is assumed.	Your stage, symptoms and treatment determine follow-up. Write down the skin and lymph-node checks, any body or brain scans and the service arranging them. Confirm your own dates with the team.

Care / check	Before starting	During treatment	Between visits	Response / follow-up	After a hold or stop
Medicines and continuing precautions NCCP	Review all medicines and the pregnancy/ fertility plan.	Review treatment taken or received, holds, adverse effects and supportive medicines.	Keep the actual urgent contacts and do not restart a held medicine yourself.	Bring the medicine list and treatment summary to other services.	Discuss fertility before starting. Use effective contraception during treatment and for at least 4 months after the last dose. Tell the team promptly about a possible pregnancy. Discuss breastfeeding restrictions with the team before treatment; do not breastfeed without their advice. Do not stop prescribed steroids or hormone replacement yourself.

What each check is for

1

Ongoing treatment and review

Every 3 weeks

Pembrolizumab helps immune cells act against melanoma that cannot be completely removed or has spread. The aim is disease control; benefit and side effects are reviewed for you.

Ask / **act:** Confirm the current phase, prescription, tests and support plan.

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Practical treatment planning

Agree the arrangements with your oncology team.

STEP 1

Taking or receiving treatment

Pembrolizumab is given as 200 mg by IV infusion over 30 minutes, on Day 1 every 3 weeks. Allow extra time for assessment, preparation and observation.

STEP 2

Missed, vomited or held treatment

If a visit is missed or treatment is held, contact the unit for a new plan. Do not count a missed appointment as a dose received. An interval change, operation or new phase does not erase earlier immune toxicity or a permanent-stop decision.

STEP 3

Appointments and supply

Plan blood tests, travel and medicine collection in advance. If you need an interpreter or someone to help at home, tell the team before the appointment.

STEP 4

Paper copies and contact information

Ask for your latest schedule and contact details on paper. Replace old versions when treatment changes so everyone helping you uses the same plan.

Taking treatment and preparing for visits

Keep the written medicine plan beside this guide; it records your own dose and dates.

Taking or receiving the medicine

Other medicines and supplements

Show the pharmacist a current list of prescriptions, over-the-counter medicines, vitamins, herbal products and supplements. Include anticoagulants, heart medicines, steroids and recent antibiotics. Check before adding, stopping or changing anything; an apparently harmless product can alter treatment or confuse a symptom assessment.

Practical preparation

Bring your medicine list, allergy information and questions. Tell the team about practical problems with travel, swallowing, wound care or remembering doses before treatment starts.

Follow-up and your continuing record

Keep this record when treatment is paused, changed or completed.

Your follow-up plan

Your stage, symptoms and treatment determine follow-up. Write down the skin and lymph-node checks, any body or brain scans and the service arranging them. Confirm your own dates with the team.

Continuing medicine precautions

Continue to report possible immune effects after the last dose; your team records the follow-up tests and dates. Keep any prescribed steroid taper, hormone replacement and specialist follow-up. Scans follow your individual disease plan.

Treatment summary for other clinicians

Record the medicines and last dates, important reactions, any long-term hormone replacement or steroid taper, and the responsible oncology contact. Show this information in urgent care and to other specialists. A different treatment phase does not erase earlier toxicity.

The complete paper version

Use this handbook as your paper record. Keep the current schedule, medicines, tests and contacts together, and replace outdated pages after a change. The matching QR link is optional.

CHAPTER 03

Symptoms and supportive care

IN THIS CHAPTER

Recognising changes, knowing when to call, and using your prescribed supportive-care plan.

Day 1 → Day 21: a general pattern

This is a visual orientation, not a prediction of your personal cycle. It shows when some effects often emerge and which effects are **not** tied to a particular day.

Treatment or check	Timing in this view
Pembrolizumab treatment	 **Day 1** Pembrolizumab is given as 200 mg by IV infusion over 30 minutes, on Day 1 every 3 weeks. Allow extra time for assessment, preparation and observation.
Symptoms between visits	 **Days 1–21** Immune effects can begin at any time, including after treatment stops.
Prepare next-cycle review	 **Day 21** Arrange the required bloods and symptom review before the next dose. Thyroid checks follow their own interval.

Your schedule: Pembrolizumab every 3 weeks. Treatment continues until progression or unacceptable toxicity. For sustained response without progression, stopping after 24 months may be considered by the treating team; this is not an automatic stop.

Tests between visits: Blood count, kidney/liver tests and glucose before every 3-week cycle. TSH is checked every 3–6 weeks, including between treatment visits when needed. Cortisol and other tests are added when clinically indicated. Report new symptoms at every visit and between visits; normal blood results do not rule out immune toxicity.

Held or missed treatment: If a visit is missed or treatment is held, contact the unit for a new plan. Do not count a missed appointment as a dose received. An interval change, operation or new phase does not erase earlier immune toxicity or a permanent-stop decision.

Safety decisions: Routine dose reductions are not recommended for these immunotherapies. The clinician may hold or stop treatment and treat the immune effect.

> **Why these bands stay broad:** This calendar shows the pattern. Your team supplies actual dates, prescribed doses and any hold or restart instructions. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00455 v16: current regimen source ↗

There are no guaranteed "safe days". The prescribed dose, treatment decision and local urgent-contact plan take priority. Call 112 or 999 for severe breathing difficulty, severe chest pain, collapse, a new seizure, sudden weakness or speech difficulty, or uncontrolled bleeding. Follow your oncology unit's written urgent-contact plan for other new or worsening symptoms; do not wait for the next appointment.

Between-treatment notes

Day 1: Treatment visit

Pembrolizumab is given as 200 mg by IV infusion over 30 minutes, on Day 1 every 3 weeks. Allow extra time for assessment, preparation and observation. Report symptoms and previous reactions before treatment.

Day 2–21: Between visits

Immunotherapy can cause inflammation in healthy organs during treatment or after it stops. Contact the oncology team promptly for new diarrhoea, blood in stools, cough or breathlessness, yellow skin or eyes, severe headache, vision changes, marked thirst, unusual weakness or a spreading rash. Tell any clinician that you are receiving or have received immunotherapy. Do not wait for the next treatment to report a possible immune effect.

Support around the medicines

1 An individual symptom plan

Ask which anti-sickness or other symptom medicines are prescribed, when to take them and when to call if they do not help. Do not start a borrowed medicine or alter treatment to manage a side effect yourself.

2 Your urgent contacts

Keep your actual oncology, out-of-hours and pharmacy contacts on the wallet card. Example numbers on a template must be replaced before you use it.

3 Pregnancy and fertility precautions

Discuss fertility before starting. Use effective contraception during treatment and for at least 4 months after the last dose. Tell the team promptly about a possible pregnancy. Discuss breastfeeding restrictions with the team before treatment; do not breastfeed without their advice. Do not stop prescribed steroids or hormone replacement yourself.

4 Eating, activity and practical help

Let the team know if cramps, altered taste or poor appetite make eating or moving difficult. Ask for practical food and activity advice that fits your symptoms and any other illness.

5 Steroids and hormone replacement

Some immune effects need steroids or lasting hormone replacement. Follow the exact prescription and taper. Carry the relevant alert information, and do not stop or reduce these medicines yourself when anticancer treatment is held or completed.

Tests: what is required and what each result means

Safety tests, tumour tests and response scans answer different questions.

Before treatment

Blood count, kidney and liver tests, glucose and thyroid tests; review autoimmune conditions, transplants, previous immune reactions and all current medicines.

During treatment

Blood count, kidney/liver tests and glucose before every 3-week cycle. TSH is checked every 3–6 weeks, including between treatment visits when needed. Cortisol and other tests are added when clinically indicated. Report new symptoms at every visit and between visits; normal blood results do not rule out immune toxicity.

Scans and tumour markers

Ask the team to write the next assessment and who will arrange it. Bloods used to monitor kidneys, liver, hormones or muscle are not automatically measures of cancer response. There is no universal tumour-marker blood schedule for these skin-cancer regimens; a scan or extra blood test is requested for a specific clinical reason.

Getting the result

Keep a named contact for bookings and results. If an expected appointment or result is missing, call that service; do not take silence as confirmation that everything is complete.

What to expect between visits

You can feel well and still need the planned checks. New symptoms deserve attention even on a day without an appointment.

Your usual pattern

Record the start and change in symptoms, temperature if advised, bowel pattern, drinking, urine and the effect on normal tasks. A simple few-line record is enough. Use it to describe a change rather than trying to grade or diagnose a toxicity yourself.

Eating, drinking and activity

Report nausea, mouth discomfort, diarrhoea or weight loss before they make eating and drinking difficult. The team can adjust symptom support and arrange dietitian or rehabilitation advice.

A held dose or changed plan

If a visit is missed or treatment is held, contact the unit for a new plan. Do not count a missed appointment as a dose received. An interval change, operation or new phase does not erase earlier immune toxicity or a permanent-stop decision. Feeling better or having normal tests does not by itself mean that pembrolizumab can restart. Your team reviews what happened, recovery and the steroid or hormone plan. Some reactions mean that treatment must remain stopped, even if the symptoms have settled. Follow the prescribed plan and never restart treatment or change steroids yourself. Before any restart, your team also reviews the original event, recovery and any steroid or hormone plan. Follow the prescribed plan; do not alter steroids or restart treatment yourself.

When heart, muscle and nerve symptoms overlap

Heart, muscle and nerve problems can occur together. New muscle pain or weakness with changed vision, unusual tiredness, breathlessness, chest pain or an irregular heartbeat needs immediate medical advice. Do not wait for several symptoms to appear together. Severe breathing difficulty, collapse or rapidly worsening weakness needs emergency care.

Skin checks, sun protection and optional photographs

This complements clinical assessment and does not replace it.

Protect your skin

Protect exposed skin with clothing, a hat and shade. Use broad-spectrum sunscreen with sun protection factor (SPF) 30 or higher and reapply as directed. Avoid sunbeds. Cloudy weather does not remove ultraviolet exposure.

Know the pattern of your skin

Ask the team to show you how to check your skin and treated area. Use a mirror or a helper for areas you cannot see. Report a new lump, a changing lesion or a sore that does not heal.

Photographs are optional

A private, dated photograph can help describe a change if your team finds it useful. Keep the lighting and size reference consistent. Photographs support a discussion; they do not replace examination and need not be uploaded.

Wounds, nails and sensitive skin

Follow the wound and skin-care instructions for the treated area. Ask before adding a cream or remedy. Increasing redness, discharge, pain, bleeding or difficulty using the area needs assessment.

Pregnancy, fertility and protecting others

Discuss these precautions before treatment even if pregnancy is not currently planned.

Medicine-specific precautions

Planning ahead

Ask early about fertility preservation, sexual health and symptoms affecting intimacy. Tell the team about any chance of pregnancy, breastfeeding or a partner's pregnancy so that the correct medicine-specific advice can be given. Keep the precautions and their end dates in the written plan.

Safe handling and supply

Do not share anticancer medicines. Keep oral medicines in their original labelled packaging, away from children and anyone who could take them accidentally. Return unused medicines to the supplying pharmacy and follow its handling advice. A treatment hold does not remove pregnancy precautions.

Questions worth asking

What tests do I need?

Blood count, kidney and liver tests, glucose and thyroid tests; review autoimmune conditions, transplants, previous immune reactions and all current medicines. Blood count, kidney/liver tests and glucose before every 3-week cycle. TSH is checked every 3–6 weeks, including between treatment visits when needed. Cortisol and other tests are added when clinically indicated. Report new symptoms at every visit and between visits; normal blood results do not rule out immune toxicity.

Who do I call between visits?

Ask the team to enter actual daytime and out-of-hours numbers on the wallet card. Report new symptoms promptly; do not wait for the next appointment.

Try explaining the plan back: Can you explain your treatment setting, medicine schedule, next tests, the symptoms that need urgent advice and the number you would call?

Your diagnosis and treatment setting Unresectable or metastatic melanoma in adults; the team confirms the source-specific eligibility and previous treatment.

Supportive prescription draft for review. This is a regimen template. It is not an issued patient prescription; individual review and completion are required.

Ongoing treatment and review

Active medicines: Pembrolizumab

Emetogenic risk: Minimal

No routine prophylactic antiemetic or premedication is specified for this monotherapy. Any symptom-treatment medicine requires an individual authorised prescription.

Additional phase-specific support

- Immunotherapy can cause inflammation in healthy organs during treatment or after it stops. Contact the oncology team promptly for new diarrhoea, blood in stools, cough or breathlessness, yellow skin or eyes, severe headache, vision changes, marked thirst, unusual weakness or a spreading rash. Tell any clinician that you are receiving or have received immunotherapy.
- Discuss fertility before starting. Use effective contraception during treatment and for at least 4 months after the last dose. Tell the team promptly about a possible pregnancy. Discuss breastfeeding restrictions with the team before treatment; do not breastfeed without their advice. Do not stop prescribed steroids or hormone replacement yourself.
- Immune adverse effects may require corticosteroids and/or long-term hormone replacement under specialist direction. Do not stop prescribed treatment yourself.
- This is an incomplete clinical/pharmacy review draft, not an issued prescription.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

What to watch for

You do not need to know which medicine caused a symptom before calling.

1 Hormones / head / eyes

Pembrolizumab

New severe headache, visual change, unusual tiredness, dizziness, thirst or frequent urination may signal immune effects and needs prompt advice.

3 Heart / muscles

Pembrolizumab

Chest pain, palpitations, marked muscle weakness or swallowing difficulty can be serious immune effects. Seek urgent assessment.

5 Kidneys / urine

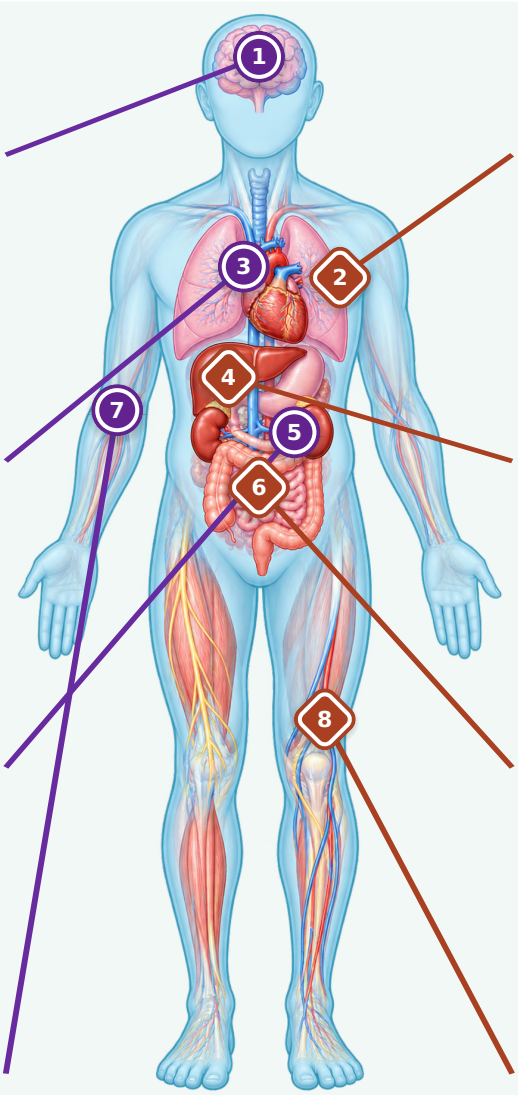
Pembrolizumab

Kidney inflammation may be detected on blood tests. Report reduced urine, swelling, blood in urine or a marked change from normal.

7 Skin / mouth

Pembrolizumab

Report rash or itching early. Blistering, peeling, mouth or genital sores, or rash with fever needs urgent assessment.



2 Lungs / breathing

Pembrolizumab

New cough or breathlessness may reflect lung inflammation or another serious cause. Call promptly; severe breathing difficulty is an emergency.

4 Liver

Pembrolizumab

Inflammation may first show in blood tests. Report yellow eyes or skin, dark urine, unusual itching or upper-abdominal pain.

6 Bowels / abdomen

Pembrolizumab

Report increased stool frequency, diarrhoea, blood or mucus, or abdominal pain early. Severe pain, black stools or dehydration needs urgent help.

8 Nerves / strength

Pembrolizumab

New tingling, weakness, unsteadiness, facial change or trouble swallowing needs prompt review. Sudden weakness or speech difficulty is an emergency.

Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

Immunotherapy can cause inflammation in healthy organs during treatment or after it stops. Contact the oncology team promptly for new diarrhoea, blood in stools, cough or breathlessness, yellow skin or eyes, severe headache, vision changes, marked thirst, unusual weakness or a spreading rash. Tell any clinician that you are receiving or have received immunotherapy.

When to get help

Call 112 / 999 for severe breathing difficulty, severe chest pain, collapse or a seizure.

Call oncology immediately: temperature 37.5°C or above, 36°C or below, shivering or suddenly feeling unwell, even with a normal temperature. Follow your hospital's temperature instructions. If you cannot reach the team, seek urgent medical assessment. Do not wait for the next appointment.

Oncology daytime: 087 12345678 · **Out of hours:** 087 87654321

EXAMPLE ONLY — team must replace both numbers before patient use.

CHAPTER 04

Sources and further information

IN THIS CHAPTER

The references behind this information and addresses you can use to find further help.

Use this information alongside the plan agreed with your treatment team. Your team confirms your medicines, doses and appointments.

1. **National Cancer Control Programme (NCCP) 00455 v16: current regimen source**
   https://healthservice.hse.ie/documents/6903/Pembrolizumab_200mg_Monotherapy_455.pdf
2. **EMA: Keytruda medicine overview**
   https://www.ema.europa.eu/en/medicines/human/EPAR/keytruda
3. **NCI: melanoma — patient information**
   https://www.cancer.gov/types/skin/patient/melanoma-treatment-pdq
4. **NCI: melanoma treatment — clinical evidence**
   https://www.cancer.gov/types/skin/hp/melanoma-treatment-pdq
5. **NCI: eating and nutrition during cancer treatment**
   https://www.cancer.gov/about-cancer/treatment/side-effects/nutrition
6. **NCI: palliative care during cancer treatment**
   https://www.cancer.gov/about-cancer/advanced-cancer/care-choices/palliative-care-fact-sheet
7. **NCI: skin cancer — patient information**
   https://www.cancer.gov/types/skin/patient/skin-treatment-pdq
8. **HSE SunSmart: protection from sun and sunbeds**
   https://www.hse.ie/conditions/cancer/protect-skin-sun-sunbeds/
9. **EMA Keytruda intravenous (IV) product information and patient leaflet**
   https://www.ema.europa.eu/en/documents/product-information/keytruda-epar-product-information_en.pdf